DETAILED DESCRIPTION OF THE INVENTION

The following description is provided to enable any person skilled in the art to make and use the disclosed subject matter and is presented in the context of particular embodiments. Various modifications, substitutions, and variations will be apparent to those skilled in the art and may be made without departing from the scope of the disclosure. Accordingly, the disclosure is not intended to be limited to the embodiments described herein.

In general, the disclosed device (FIG. 4A) is an intraluminal compliance and distance profiling instrument comprising an insertable measurement section, a distal sensing tip configured to acquire a distance measurement to an internal anatomical boundary, a compliance characterization region configured to measure tissue response, a positional reference element configured to establish repeatable insertion depth, and a handle portion configured to support user manipulation and house electronics and/or pumping components.

In various embodiments, the device is configured for use within vaginal, anal, or other intraluminal anatomical environments and may be utilized as a standalone measurement device or as a sensing node within a multi-device physiologic measurement platform. The device may be provided in multiple model tiers, including simplified embodiments emphasizing minimal component count and advanced embodiments emphasizing enhanced accuracy, robustness, or geometry reconstruction capability.

The insertable measurement section may have an effective insertable length in the range of approximately 2 to 7 inches, and in certain preferred embodiments approximately 3.33 inches. The insertable section may be generally cylindrical, tapered, or otherwise shaped to promote comfort and stable positioning. The internal backbone may be soft and conformable, semi-rigid with a compliant overmold, or segmented to permit safe bending while maintaining alignment between sensing regions.

The distal sensing tip is configured to acquire distance to an anatomical terminus using one or more ranging modalities including optical time-of-flight sensing, LiDAR-based time-of-flight sensing, ultrasonic time-of-flight sensing, or combinations thereof (FIG. 4B). In simplified embodiments, a single ranging sensor provides a scalar distance measurement processed by a minimal time-of-flight computation. In advanced embodiments, the distal region may acquire multiple samples across time or orientation to improve robustness or to support coarse geometry characterization.

The compliance characterization region may comprise one or more pressure sensing elements and/or an inflatable expansion structure (FIGS. 4C–4E). In pressure-based embodiments, one or more sensors may be positioned circumferentially and/or longitudinally along the insertable section to detect contact pressure and generate a pressure response profile. In inflatable embodiments, an expansion region may be inflated to controlled levels and measured using pressure sensing, strain sensing, pump telemetry, or combinations thereof to generate compliance curves describing expansion behavior over time.

A positional reference element (FIG. 4F) may be located between the insertable section and the handle to provide a repeatable insertion-depth limit. The reference element may include a flange, stop, compliant barrier, or ergonomic transition and may be configured to improve repeatability, reduce user error, and enhance safety by limiting over-insertion.

The handle portion (FIG. 4G) may include a grip region and may house power components, processing circuitry, wireless communication modules, optional inertial sensing, optional user feedback elements, and, in inflatable embodiments, a manual pump and/or automated pump. The handle may be sized to be as compact as practical while maintaining ease of use and stable manipulation during measurement sessions.